YANHUI YANG

yyang440@ucr.edu ⋄ https://astro-yyh.github.io

EDUCATION

PhD Program in Physics University of California, Riverside — Sep. 2022–current
Concentration in Astronomy
Overall GPA: 4.00/4.00

BS in Astronomy University of Science and Technology of China — Jul. 2021
Wang Shouguan Talent Program in Astronomy
Thesis: *Hydrodynamic simulations of turbulent mixing layers of galactic winds*
Overall GPA: 3.74/4.30, Major GPA: 3.93/4.30 (Physics & Astronomy courses)

RESEARCH INTERESTS

Cosmology: large-scale-structure, dark matter, dark energy, cosmological emulation; Astrophysics: galaxy formation and evolution, circumgalactic medium, turbulent mixing layers

APPOINTMENTS

Graduate Student Researcher, Advisor: Simeon Bird (UCR)	Fall 2023–current
Teaching assistant in *General Physics laboratory 2LC*, Coordinator: Jonathan Eldridge (UCR)	Winter–Spring 2023
Research assistant, Advisor: Suoqing Ji (SHAO)	Spring 2022

ADVISORS

Simeon Bird (sbird@ucr.edu)
Associate Professor, Department of Physics and Astronomy, University of California, Riverside

Suoqing Ji (sqji@fudan.edu.cn)
Associate Professor, Department of Physics, Fudan University

Philip F. Hopkins (phopkins@caltech.edu)
Professor, The Division of Physics, Mathematics and Astronomy, California Institute of Technology

Xiaobo Dong (xbdong@ynao.ac.cn)
Research Professor, Yunnan Observatories, Chinese Academy of Sciences

MENTORSHIP & SUPERVISION

Undergraduate students (co-mentored with Prof. Simeon Bird): Mark Achenbach (UH Mānoa), Michael Padilla (UCR), Mohak Bhattacharya (UCR)

SKILLS

Programming languages	C (gsl), C++ (boost, CUDA), Python (torch, mpi4py, pymc, pandas, numpy, matplotlib, yt, bigfile, gpy, etc.), Bash, Fortran, HTML, CSS, Markdown, LaTeX
Simulation codes	MP-Gadget, CLASS, FLASH, GADGET-3, RAMSES, AREPO
Tools	Git, Mathematica, Origin, Gnuplot, IDL, ds9, VS Code

VISITS & ADDITIONAL TRAINING

TACC Open Hackathon, Texas Advanced Computing Center, UT Austin — Oct. 2024

The 2023 Machine Learning Institute, TACC, UT Austin — Jun. 2023

Shanghai Astronomical Observatory, Chinese Academy of Sciences — Jan. 2020

Five-hundred-meter Aperture Spherical Telescope (FAST) — Jul. 2019

AWARDS & HONORS

Dean's Distinguished Fellowship (University of California, Riverside) — 2022

NAOC Scholarship (National Astronomical Observatories, Chinese Academy of Sciences) — 2020

National Encouragement Scholarship — 2020

Outstanding Student Scholarship (Grade 3) — 2019

Zhang Zongzhi Sci-Tech Scholarship — 2018

PROFESSIONAL SERVICE

Journal referee for *Physical Review D* and *Physical Review Letters*

PRESENTATIONS

Invited Talks

Modeling Cosmic Large-Scale Structure Beyond ΛCDM with Simulations and Emulation
Fudan-CfAA Seminar, Fudan University — Oct. 2025

Goku: A 10-parameter simulation suite for cosmological emulation
Physics and Astronomy Student Seminar, University of California, Riverside — Mar. 2025

Radiative turbulent mixing layers at high Mach numbers
Shanghai Astronomical Observatory, Chinese Academy of Sciences — May 2022

Contributed Talks/Posters

Cosmological Emulation based on the Goku Simulations in a 10-Dimensional Parameter Space
XIX International Conference on Interconnections between Particle Physics and Cosmology (PPC2026)
The University of New South Wales, Sydney — Aug. 2026

Improving Cosmological Emulation via Multifidelity Neural Networks
Galaxy Formation & Evolution in Southern California, University of California, San Diego — Sep. 2025

Nonlinear Matter Power Spectrum Emulation beyond ΛCDM for Next-Generation Surveys (poster)
Cosmic Cartography with Roman 2025, STScI, Baltimore, MD — Jul. 2025

10D emulation for the nonlinear matter power spectrum beyond ΛCDM based on Goku
Cosmology from Home 2025 — Jul. 2025

Goku: A 10-parameter simulation suite for cosmological emulation
Frontera User Meeting 2024, TACC, University of Texas, Austin — Aug. 2024

OUTREACHES

Physics Fun Expedition: A Science Outreach Program for Township Middle Schools — Jul. 2019

PUBLICATIONS

Yanhui Yang, Simeon Bird, Yihao Zhou, Tiziana Di Matteo, Rupert Croft, Yueying Ni, and Nianyi Chen. Matter Clustering in Astrid: Reduced Baryonic Suppression from Realistic Black Hole Dynamics. *ApJL*, 1007(1):L20, August 2026. doi: 10.3847/2041-8213/ae8dc6.

Yanhui Yang, Simeon Bird, Ming-Feng Ho, and Mahdi Qezlou. Ten-dimensional neural network emulator for the nonlinear matter power spectrum. *Phys. Rev. Lett.*, 136(6):061001, February 2026a. doi: 10.1103/8yzy-t59w.

Yanhui Yang, Simeon Bird, Ming-Feng Ho, and Mahdi Qezlou. Design and optimization of neural networks for multifidelity cosmological emulation. *Phys. Rev. D*, 113(4):043508, February 2026b. doi: 10.1103/cqrc-k8wq.

Yanhui Yang, Simeon Bird, and Ming-Feng Ho. Ten-parameter simulation suite for cosmological emulation beyond ΛCDM. *Phys. Rev. D*, 111(8):083529, April 2025. doi: 10.1103/PhysRevD.111.083529.

Yanhui Yang and Suoqing Ji. Radiative turbulent mixing layers at high Mach numbers. *MNRAS*, 520(2):2148–2162, April 2023. doi: 10.1093/mnras/stad264.

Only lead-author papers are listed above. Please refer to NASA/ADS (orcid:0000-0001-6221-6024) for a complete list of my publications.